

# Chapter 8: Area and volume

13 lessons · one lesson for each exercise · each lesson is one hour

**NSW syllabus outcome**

Stage 4 Measurement and space: MA4-LEN-C-01 (length and circumference), MA4-ARE-C-01 (area) and MA4-VOL-C-01 (volume and capacity).

CHAPTER

8

**8.01** Units of measurement

**8.02** Perimeter

**8.03** Circumference of a circle

**8.04** Area units

**8.05** Area of squares and rectangles

**8.06** Area of a triangle

**8.07** Area of a parallelogram

**8.08** Area of composite shapes

**8.09** Prisms, cross-sections and views

**8.10** Volume units

**8.11** Volume of a rectangular prism

**8.12** Volume of a prism

**8.13** Capacity

## How to use this booklet

**1** Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

### WE DO

**Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

**3** Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

**2** Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

### YOU DO

**Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

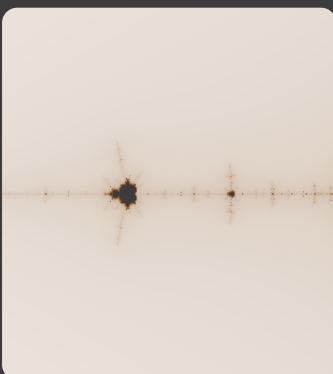
**4** Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

## Contents and progress tracker

| Lesson      | Topic                            | Page      | Date | Done                     |
|-------------|----------------------------------|-----------|------|--------------------------|
| <b>8.01</b> | Units of measurement             | <b>3</b>  |      | <input type="checkbox"/> |
| <b>8.02</b> | Perimeter                        | <b>7</b>  |      | <input type="checkbox"/> |
| <b>8.03</b> | Circumference of a circle        | <b>13</b> |      | <input type="checkbox"/> |
| <b>8.04</b> | Area units                       | <b>19</b> |      | <input type="checkbox"/> |
| <b>8.05</b> | Area of squares and rectangles   | <b>23</b> |      | <input type="checkbox"/> |
| <b>8.06</b> | Area of a triangle               | <b>29</b> |      | <input type="checkbox"/> |
| <b>8.07</b> | Area of a parallelogram          | <b>35</b> |      | <input type="checkbox"/> |
| <b>8.08</b> | Area of composite shapes         | <b>41</b> |      | <input type="checkbox"/> |
| <b>8.09</b> | Prisms, cross-sections and views | <b>47</b> |      | <input type="checkbox"/> |
| <b>8.10</b> | Volume units                     | <b>53</b> |      | <input type="checkbox"/> |
| <b>8.11</b> | Volume of a rectangular prism    | <b>57</b> |      | <input type="checkbox"/> |
| <b>8.12</b> | Volume of a prism                | <b>63</b> |      | <input type="checkbox"/> |
| <b>8.13</b> | Capacity                         | <b>69</b> |      | <input type="checkbox"/> |

## In this chapter you will

- convert units of length, mass, capacity and time
- find perimeter and circumference
- convert area units and find the area of rectangles, triangles, parallelograms and composite shapes
- draw cross-sections and views of prisms
- convert volume units and find the volume of prisms
- find capacity in litres



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON  
**8.01**

# Units of measurement

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- convert between units of length, mass, capacity and time

## Success criteria

I can:

- ☐ know the metric prefixes: kilo, centi, milli
- ☐ multiply to convert to a smaller unit, divide to convert to a larger unit
- ☐ convert units of time

**Key terms** Write the definition as your teacher explains it.

**Metric**

**Convert**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Convert. Bigger unit to smaller: multiply.

**a** Convert 3200 m to km.

**b** Convert 0.042 kg to g.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Convert.

14 QUESTIONS

**REMEMBER**  $\text{km} \rightarrow \text{m} \times 1000$ ,  $\text{m} \rightarrow \text{cm} \times 100$ ,  $\text{cm} \rightarrow \text{mm} \times 10$ .

**1** 3200 m to km

**2** 904 cm to mm

**3** 7150 mm to m

**4** 0.042 kg to g

**5** 3.5 kg to t

**6** 125 L to kL

Questions 7–14: convert.

**7** 5 km to m

**8** 350 cm to m

**9** 2.4 kg to g

**10** 1500 mL to L

**11** 3 h to min

**12** 90 s to min

**13** 6 cm to mm

**14** 4000 kg to t

**YOU DO** On your own: **Easy** Set B. Convert.

6 QUESTIONS

1 8.3 L to mL

4 15 h to min

2 4.2 kL to mL

5 2.5 days to h

3 240 min to h

6 17 000 000 mg to kg

**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Convert capacity and time.

**a** Convert 8.3 L to mL.

Your teacher works this out. Copy every step.

**b** Convert 240 min to h.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Convert. Show your working.

6 QUESTIONS

**REMEMBER**  $\text{kg} \rightarrow \text{g} \times 1000$ ,  $\text{t} \rightarrow \text{kg} \times 1000$ ,  $\text{L} \rightarrow \text{mL} \times 1000$ .

1 15 h to s

3 4500 g to kg

5 750 mL to L

2 0.65 km to cm

4 3 h 20 min to min

6 1 week to min

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Convert, then solve.

**a** Convert 15 h to seconds.

Your teacher works this out. Copy every step.

**b** How many minutes has Isla lived by her 12th birthday? (365 days a year)

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Bigger unit → smaller unit: multiply. Smaller → bigger: divide.

**1** How many minutes has Isla lived by her 12th birthday? (365 days a year)

**3** A tap drips 5 mL a minute. How many litres in one day?

**2** A recipe needs 350 g of flour for one cake. How many kg for 12 cakes?

**4** Order from smallest to largest: 1.2 km, 950 m, 125 000 cm, 1 100 000 mm.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Convert 4.5 m to cm.

**Level 2**

Convert 2750 g to kg.

**Level 3**

Convert 3.5 h to minutes.

 Extension · Stretch yourself Worked example

Light travels 300 000 km in one second. How far does it travel in one hour?

1.  $1 \text{ hour} = 3600 \text{ s}$

2.  $300\,000 \times 3600$

Answer: 1 080 000 000 km

## YOU DO

On your own: read the worked example, then try these.

- E1 Your heart beats about 70 times a minute. About how many times does it beat in a year?

- E2 Convert 1 million seconds to days (1 d.p.).

## Lesson summary

Missed the lesson? Start here.

## Conversions

1.  $\text{km} \rightarrow \text{m} \times 1000$ ,  $\text{m} \rightarrow \text{cm} \times 100$ ,  $\text{cm} \rightarrow \text{mm} \times 10$ .

2.  $\text{kg} \rightarrow \text{g} \times 1000$ ,  $\text{t} \rightarrow \text{kg} \times 1000$ ,  $\text{L} \rightarrow \text{mL} \times 1000$ .

3. Bigger unit  $\rightarrow$  smaller unit: multiply. Smaller  $\rightarrow$  bigger: divide.

4. Time:  $60 \text{ s} = 1 \text{ min}$ ,  $60 \text{ min} = 1 \text{ h}$ ,  $24 \text{ h} = 1 \text{ day}$ .

 Level 1

3200 m to km

1.  $\div 1000$

Answer: 3.2 km

 Level 2

8.3 L to mL

1.  $\times 1000$

Answer: 8300 mL

 Level 3

15 h to s

1.  $15 \times 60 \times 60$

Answer: 54 000 s

 Extension

Light in 1 hour

1.  $300\,000 \times 3600$

Answer: 1 080 000 000 km

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
8.02

# Perimeter

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the perimeter of shapes

## Success criteria

I can:

- ☐ add all the side lengths
- ☐ find missing side lengths first
- ☐ use the same units for every side

**Key terms** Write the definition as your teacher explains it.

**Perimeter**

**Composite shape**

Notes

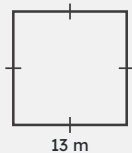
## Level 1 · Getting started

Easy · 15 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the perimeter.

**a** Find the perimeter of the square.



**b** Find the perimeter.



Your teacher works this out. Copy every step.

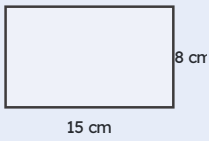
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Find the perimeter.

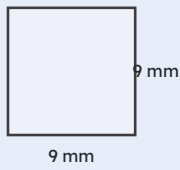
6 QUESTIONS

**REMEMBER** Perimeter = distance around the outside.

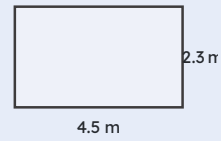
1



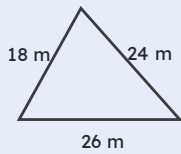

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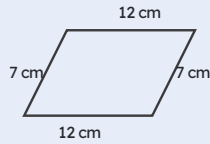

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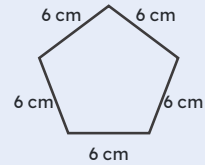

2




4




6




**YOU DO** On your own: **Easy** Set A2. Quick drill: find the perimeter.

6 QUESTIONS

1 square, side 7 cm

2 rectangle 8 m by 3 m

3 equilateral triangle, side 5 mm

4 rectangle 12 cm by 4.5 cm

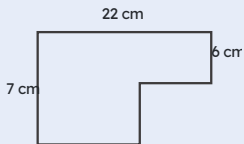
5 regular hexagon, side 6 m

6 square, side 2.5 km

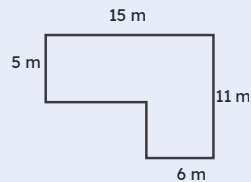
**YOU DO** On your own: **Easy** Set B. Find the perimeter.

3 QUESTIONS

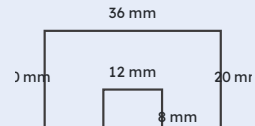
1




2




3



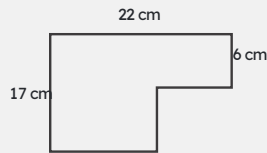
Level 2 · Building up

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Find the perimeter. Find missing sides first.

**a** Find the perimeter.



Your teacher works this out. Copy every step.

**b** Find the perimeter of a rectangle 2.4 m by 85 cm (in cm).

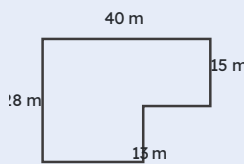
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the perimeter. Find missing sides first.

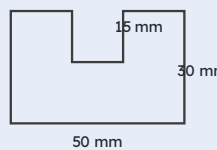
6 QUESTIONS

**REMEMBER** Add every side length.

1



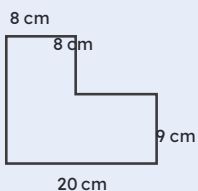
3



5

A square has perimeter 46 cm. Find its side.

2



4

A rectangle is 2.4 m by 85 cm. Find its perimeter in cm.

6

A rectangle has perimeter 30 m and length 9 m. Find its width.

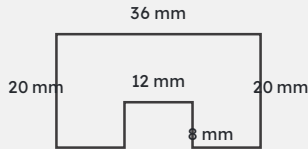
Level 3 · Challenge

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the perimeter, then solve.

**a** Find the perimeter.



Your teacher works this out. Copy every step.

**b** A square has perimeter 46 cm. Find its side length.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Find missing sides using opposite sides.

**1** Fencing costs \$24.50 per metre. Find the cost to fence a 35 m by 18 m paddock.

**3** How many different rectangles with whole-number sides have a perimeter of 20 cm?

**2** A regular hexagon has perimeter 57 cm. Find the length of one side.

**4** A photo 15 cm by 10 cm has a 2 cm frame around it. Find the outside perimeter of the frame.



## Extension · Stretch yourself

Finished Set D? Stretch yourself here.



## Worked example

A rectangle's length is 3 times its width. Its perimeter is 64 cm. Find its dimensions.

1. Width  $w$ , length  $3w$
2.  $2(w + 3w) = 8w = 64$
3.  $w = 8$

**Answer:** 8 cm by 24 cm

## YOU DO

On your own: read the worked example, then try these.

- E1** A rectangle's length is 5 cm more than its width. Its perimeter is 50 cm. Find its dimensions.

- E2** An isosceles triangle has perimeter 40 cm and base 12 cm. Find the equal sides.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

## YOU DO

## Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the perimeter of a rectangle 12 cm by 7 cm.

**Level 2**

Find the perimeter of an equilateral triangle with sides 9.5 m.

**Level 3**

A square has perimeter 60 mm. Find its side.

## Lesson summary

Missed the lesson? Start here.

### Remember

1. Perimeter = distance around the outside.
2. Add every side length.
3. Find missing sides using opposite sides.
4. Check all units are the same.

#### Level 1

Square, side 13 m

1.  $4 \times 13$

Answer: 52 m

#### Level 2

Rectangle 13 mm  $\times$  5 mm

1.  $2 \times (13 + 5)$

Answer: 36 mm

#### Level 3

L-shape 22, 6, 17...

1. Missing sides:  $22 - 10 = 12$ ,  $17 - 6 = 11$

Answer: 78 cm

#### Extension

Length =  $3 \times$  width,  $P = 64$

1.  $8w = 64$

Answer: 8 cm by 24 cm

My reflection: what did I learn today? Which question was hardest, and why?

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## LESSON 8.03

## Circumference of a circle

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the circumference of a circle

### Success criteria

I can:

- ☐ name the parts of a circle: centre, radius, diameter, circumference
- ☐ use  $C = \pi d$  or  $C = 2\pi r$
- ☐ round the answer to 1 decimal place

**Key terms** Write the definition as your teacher explains it.

## Circumference

## Radius

## Diameter

## Notes

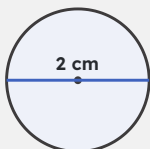
## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the circumference (1 decimal place).

**a** Find the circumference (1 d.p.).



**b** Find the circumference (1 d.p.).



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Easy** Set A. Questions 1–6: Find the diameter of each circle.

12 QUESTIONS

**REMEMBER** Diameter =  $2 \times$  radius.

1 radius 5 cm

2 radius 12 m

3 radius 3.5 mm

4 radius 0.8 km

5 radius 21 cm

6 radius 9 m

Questions 7–12: find the circumference,  $C = \pi d$  (1 d.p.).

7  $d = 4$  cm

8  $d = 10$  m

9  $r = 3$  cm

10  $r = 5$  mm

11  $d = 1$  m

12  $r = 20$  cm

**YOU DO**

On your own: **Easy** Set B. Find the circumference,  $C = \pi d$  (1 d.p.).

6 QUESTIONS

1  $d = 2$  cm

4  $d = 30$  mm

2  $d = 14$  cm

5  $d = 1.5$  km

3  $d = 7$  m

6  $d = 100$  cm

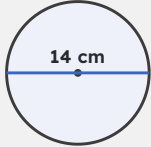
**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Find C (1 decimal place).

**a** Find C (1 d.p.).



Your teacher works this out. Copy every step.

**b** Find C (1 d.p.).



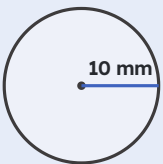
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the circumference (1 d.p.).

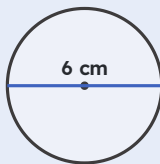
6 QUESTIONS

**REMEMBER**  $C = \pi d$  or  $C = 2\pi r$ .

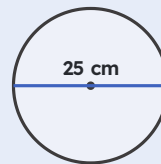
1



3



5



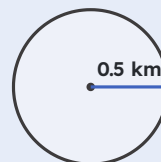
2



4



6



**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Solve.

- a** A bike wheel has diameter 66 cm. How far does it travel in 100 turns (in m)?

Your teacher works this out. Copy every step.

- b** A circle has circumference 50 cm. Find its diameter (1 d.p.).

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Use the  $\pi$  key on your calculator.

- 1** A bike wheel has diameter 66 cm. How far does it travel in 100 turns (in m, 1 d.p.)?

- 3** Find the perimeter of a semicircle with diameter 12 cm (1 d.p.).

- 2** A circle has circumference 50 cm. Find its diameter (1 d.p.).

- 4** A round table has radius 60 cm. How many 30 cm-wide chairs fit around it?

**Extension · Stretch yourself**

Finished Set D? Stretch yourself here.

**Worked example**

Find the perimeter of a quarter circle with radius 10 cm (1 d.p.).

1. Arc =  $\frac{1}{4} \times 2\pi \times 10 = 15.707\dots$
2. Add two radii:  $15.71 + 20$

**Answer:** 35.7 cm

**YOU DO**

On your own: read the worked example, then try these.

- E1** Find the perimeter of a semicircle with radius 7 m (1 d.p.).

- E2** A running track has two straights of 100 m and two semicircles of diameter 64 m. Find its length (1 d.p.).

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO**

**Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

A circle has radius 8 cm. Find its diameter.

**Level 2**

Find the circumference of a circle with diameter 9 m (1 d.p.).

**Level 3**

Find the circumference of a circle with radius 3.5 cm (1 d.p.).

## Lesson summary

Missed the lesson? Start here.

### Remember

1. Diameter =  $2 \times$  radius.
2.  $C = \pi d$  or  $C = 2\pi r$ .
3. Use the  $\pi$  key on your calculator.
4. Round to 1 decimal place and include units.

#### Level 1

$$d = 2 \text{ cm}$$

$$1. \pi \times 2 = 6.283\dots$$

Answer: 6.3 cm

#### Level 2

$$r = 10 \text{ mm}$$

$$1. 2 \times \pi \times 10$$

Answer: 62.8 mm

#### Level 3

$$C = 50, \text{ find } d$$

$$1. 50 \div \pi$$

Answer: 15.9 cm

#### Extension

$$\text{Quarter circle } r = 10$$

$$1. 15.7 + 20$$

Answer: 35.7 cm

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON  
8.04

## Area units

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- convert between units of area

## Success criteria

I can:

- ☐ know  $1 \text{ cm}^2 = 100 \text{ mm}^2$ ,  $1 \text{ m}^2 = 10\,000 \text{ cm}^2$
- ☐ multiply to convert to a smaller unit, divide to a larger unit
- ☐ choose a sensible unit of area

**Key terms** Write the definition as your teacher explains it.

Area

Square unit

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Explain, then convert.**a** Why is  $1 \text{ cm}^2 = 100 \text{ mm}^2$ ?

Your teacher works this out. Copy every step.

**b** How many  $\text{cm}^2$  in  $9.1 \text{ m}^2$ ?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Convert.

12 QUESTIONS

**REMEMBER**  $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$ .**1**  $3 \text{ cm}^2$  to  $\text{mm}^2$ 

\_\_\_\_\_

**2**  $2.5 \text{ cm}^2$  to  $\text{mm}^2$ 

\_\_\_\_\_

**3**  $9.1 \text{ m}^2$  to  $\text{cm}^2$ 

\_\_\_\_\_

**4**  $4 \text{ m}^2$  to  $\text{cm}^2$ 

\_\_\_\_\_

**5**  $2 \text{ km}^2$  to  $\text{m}^2$ 

\_\_\_\_\_

**6**  $5 \text{ ha}$  to  $\text{m}^2$ 

\_\_\_\_\_

Questions 7–12: convert.

**7**  $2 \text{ cm}^2$  to  $\text{mm}^2$ 

\_\_\_\_\_

**8**  $500 \text{ mm}^2$  to  $\text{cm}^2$ 

\_\_\_\_\_

**9**  $3 \text{ m}^2$  to  $\text{cm}^2$ 

\_\_\_\_\_

**10**  $20\,000 \text{ cm}^2$  to  $\text{m}^2$ 

\_\_\_\_\_

**11**  $4 \text{ ha}$  to  $\text{m}^2$ 

\_\_\_\_\_

**12**  $30\,000 \text{ m}^2$  to  $\text{ha}$ 

\_\_\_\_\_

**YOU DO** On your own: **Easy** Set B. Convert.

6 QUESTIONS

1 240 mm<sup>2</sup> to cm<sup>2</sup>

4 850 mm<sup>2</sup> to cm<sup>2</sup>

2 175 000 cm<sup>2</sup> to m<sup>2</sup>

5 3 500 000 m<sup>2</sup> to km<sup>2</sup>

3 50 000 m<sup>2</sup> to ha

6 6400 cm<sup>2</sup> to m<sup>2</sup>

**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Convert.

**a** How many mm<sup>2</sup> in 2.5 cm<sup>2</sup>?

Your teacher works this out. Copy every step.

**b** How many m<sup>2</sup> in 175 000 cm<sup>2</sup>?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Convert. Show your working.

6 QUESTIONS

**REMEMBER** 1 m<sup>2</sup> = 100 × 100 = 10 000 cm<sup>2</sup>.

1 240 000 mm<sup>2</sup> to m<sup>2</sup>

3 0.07 km<sup>2</sup> to ha

5 0.35 m<sup>2</sup> to cm<sup>2</sup>

2 8.6 m<sup>2</sup> to mm<sup>2</sup>

4 12.5 ha to m<sup>2</sup>

6 72 cm<sup>2</sup> to m<sup>2</sup>

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Convert.

**a** How many  $\text{m}^2$  in 240 000  $\text{mm}^2$ ?

Your teacher works this out. Copy every step.

**b** How many  $\text{mm}^2$  in 8.6  $\text{m}^2$ ?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Choose and compare.

4 QUESTIONS

**REMEMBER** 1 ha = 10 000  $\text{m}^2$ ; 1  $\text{km}^2$  = 1 000 000  $\text{m}^2$ .

**1** Which unit would you use for: a stamp, a classroom floor, a farm, a country?

**3** A tile is 400  $\text{cm}^2$ . How many tiles cover 6  $\text{m}^2$ ?

**2** Which is larger: 3.2  $\text{m}^2$  or 30 000  $\text{cm}^2$ ?

**4** A farm is 2.4  $\text{km}^2$ . How many hectares is that?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Convert 5  $\text{cm}^2$  to  $\text{mm}^2$ .

**Level 2**

Convert 30 000  $\text{cm}^2$  to  $\text{m}^2$ .

**Level 3**

Convert 1.5  $\text{m}^2$  to  $\text{mm}^2$ .

**Extension · Stretch yourself****Worked example**

Show that  $1 \text{ km}^2 = 100 \text{ ha}$ .

- $1 \text{ km}^2 = 1000 \text{ m} \times 1000 \text{ m} = 1\,000\,000 \text{ m}^2$
- $1 \text{ ha} = 10\,000 \text{ m}^2$
- $1\,000\,000 \div 10\,000$

**Answer:** 100 ha

**YOU DO**

On your own: read the worked example, then try these.

- E1** How many  $\text{mm}^2$  are in  $1 \text{ m}^2$ ?

- E2** A soccer field is 105 m by 68 m. What fraction of a hectare is it (2 d.p.)?

**Lesson summary**

Missed the lesson? Start here.

**Area conversions**

1.  $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$ .

2.  $1 \text{ m}^2 = 100 \times 100 = 10\,000 \text{ cm}^2$ .

3.  $1 \text{ ha} = 10\,000 \text{ m}^2$ ;  $1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$ .

4. To a smaller unit:  $\times$ . To a larger unit:  $\div$ .

**Level 1**

$9.1 \text{ m}^2$  to  $\text{cm}^2$

$1 \times 10\,000$

**Answer:**  $91\,000 \text{ cm}^2$

**Level 2**

$240 \text{ mm}^2$  to  $\text{cm}^2$

$1 \div 100$

**Answer:**  $2.4 \text{ cm}^2$

**Level 3**

$240\,000 \text{ mm}^2$  to  $\text{m}^2$

$1 \div 1\,000\,000$

**Answer:**  $0.24 \text{ m}^2$

**Extension**

$1 \text{ km}^2$  in ha

$1\,000\,000 \div 10\,000$

**Answer:** 100 ha

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
**8.05**

# Area of squares and rectangles

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the area of squares and rectangles

## Success criteria

I can:

- ☐ use  $A = l \times w$  for a rectangle
- ☐ use  $A = s^2$  for a square
- ☐ find a missing side from the area

**Key terms** Write the definition as your teacher explains it.

**Length**

**Width**

Notes

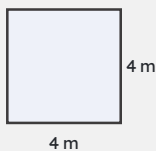
## Level 1 · Getting started

Easy · 18 questions

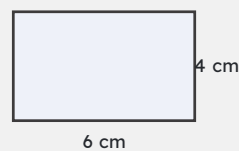
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the area.

**a** Find the area.



**b** Find the area.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Find the area.

6 QUESTIONS

**REMEMBER** Area of a rectangle = length  $\times$  width.

|                                                                                                                    |                                                                                                                     |                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <p>①</p>  <p>4 m</p>              | <p>③</p>  <p>18 m</p> <p>9 m</p>   | <p>⑤</p>  <p>7 cm</p> <p>7 cm</p>   |
| <p>②</p>  <p>6 cm</p> <p>4 cm</p> | <p>④</p>  <p>12 mm</p> <p>5 mm</p> | <p>⑥</p>  <p>2.5 m</p> <p>1.2 m</p> |

**YOU DO** On your own: **Easy** Set A2. Quick drill: find the area.

6 QUESTIONS

- |                      |                             |                           |
|----------------------|-----------------------------|---------------------------|
| ① square, side 6 cm  | ② rectangle 7 m by 4 m      | ③ rectangle 12 mm by 5 mm |
| _____                | _____                       | _____                     |
| ④ square, side 1.5 m | ⑤ rectangle 20 cm by 0.5 cm | ⑥ square, side 11 km      |
| _____                | _____                       | _____                     |

**YOU DO** On your own: **Easy** Set B. Find the area.

6 QUESTIONS

- |                                                                                                            |                                                                                                             |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <p>① rectangle 15 cm by 8 cm</p> <div style="border: 1px solid black; height: 100px; width: 400px;"></div> | <p>④ square with side 0.5 km</p> <div style="border: 1px solid black; height: 100px; width: 400px;"></div>  |
| <p>② square with side 11 m</p> <div style="border: 1px solid black; height: 100px; width: 400px;"></div>   | <p>⑤ rectangle 40 mm by 25 mm</p> <div style="border: 1px solid black; height: 100px; width: 400px;"></div> |
| <p>③ rectangle 3.5 m by 2 m</p> <div style="border: 1px solid black; height: 100px; width: 400px;"></div>  | <p>⑥ rectangle 1.2 m by 0.5 m</p> <div style="border: 1px solid black; height: 100px; width: 400px;"></div> |

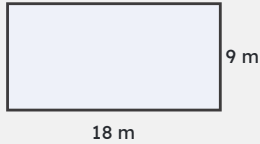
Level 2 · Building up

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Find the area. Use the same units.

**a** Find the area.



Your teacher works this out. Copy every step.

**b** A rectangle 2.5 m by 60 cm. Find its area in  $\text{m}^2$ .

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the missing side.

6 QUESTIONS

**REMEMBER** Area of a square = side  $\times$  side =  $s^2$ .

**1** A square has area  $81 \text{ cm}^2$ . Find its side.

**3** A rectangle has area  $7.5 \text{ cm}^2$  and width 2.5 cm. Find its length.

**5** A rectangle is 2.5 m by 60 cm. Find its area in  $\text{m}^2$ .

**2** A rectangle has area  $48 \text{ m}^2$  and length 8 m. Find its width.

**4** A square has area  $1.44 \text{ m}^2$ . Find its side.

**6** A square has perimeter 36 cm. Find its area.

**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the missing side.

**a** A square has area  $81 \text{ cm}^2$ . Find its side.

Your teacher works this out. Copy every step.

**b** A rectangle has area  $48 \text{ m}^2$  and length 8 m. Find its width.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Use the same units for both sides.

**1** Carpet costs \$45 per  $\text{m}^2$ . Find the cost to carpet a room 5.2 m by 4 m.

**3** A rectangle has perimeter 26 cm and area  $40 \text{ cm}^2$ . Find its length and width.

**2** How many 30 cm by 30 cm tiles cover a floor 3.6 m by 2.7 m?

**4** If you double the side of a square, what happens to its area?

**Extension · Stretch yourself**

Finished Set D? Stretch yourself here.

**Worked example**

A 20 m by 15 m lawn has a 2 m wide path around the outside of it. Find the area of the path.

1. Outer rectangle: 24 m by 19 m = 456 m<sup>2</sup>
2. Lawn: 20 × 15 = 300 m<sup>2</sup>
3. 456 – 300

**Answer:** 156 m<sup>2</sup>

**YOU DO**

On your own: read the worked example, then try these.

- E1** A 6 m by 4 m pool has a 1 m wide path around it. Find the path area.

- E2** Which rectangle with perimeter 24 cm has the largest area?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the area of a rectangle 9 cm by 6 cm.

**Level 2**

Find the area of a square with side 2.5 m.

**Level 3**

A rectangle has area 60 m<sup>2</sup> and width 5 m. Find its length.

## Lesson summary

Missed the lesson? Start here.

### Remember

1. Area of a rectangle = length  $\times$  width.
3. Use the same units for both sides.

2. Area of a square = side  $\times$  side =  $s^2$ .
4. Answer in square units.

#### Level 1

4 m square

1.  $4 \times 4$

Answer:  $16 \text{ m}^2$

#### Level 2

$18 \text{ m} \times 9 \text{ m}$

1.  $18 \times 9$

Answer:  $162 \text{ m}^2$

#### Level 3

Area 48, length 8

1.  $48 \div 8$

Answer: 6 m

#### Extension

Path around a lawn

1.  $456 - 300$

Answer:  $156 \text{ m}^2$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
8.06

# Area of a triangle

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the area of a triangle

## Success criteria

I can:

- ☐ identify the base and the perpendicular height
- ☐ use  $A = \frac{1}{2} \times b \times h$
- ☐ find the height when it is outside the triangle

**Key terms** Write the definition as your teacher explains it.

**Base**

**Perpendicular height**

Notes

## Level 1 · Getting started

Easy · 18 questions

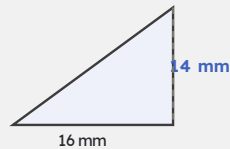
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the area.  $A = \frac{1}{2}bh$ .

**a** Find the area.



**b** Find the area.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Find the area.

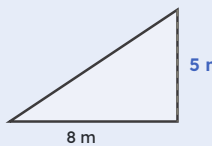
6 QUESTIONS

**REMEMBER**  $A = \frac{1}{2} \times \text{base} \times \text{height}$ .

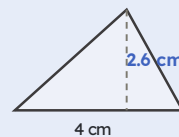
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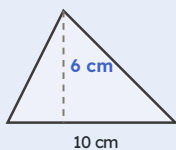

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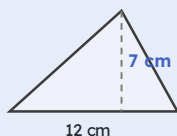

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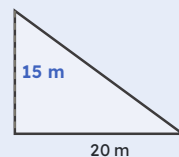

2




4




6




**YOU DO** On your own: **Easy** Set A2. Quick drill: find the area of the triangle.

6 QUESTIONS

1  $b = 10 \text{ cm}, h = 4 \text{ cm}$

2  $b = 6 \text{ m}, h = 7 \text{ m}$

3  $b = 12 \text{ mm}, h = 5 \text{ mm}$

4  $b = 8 \text{ cm}, h = 8 \text{ cm}$

5  $b = 3 \text{ m}, h = 5 \text{ m}$

6  $b = 20 \text{ km}, h = 9 \text{ km}$

**YOU DO** On your own: **Easy** Set B. Find the area of each triangle.

6 QUESTIONS

1 base 6 cm, height 4 cm

4 base 7 cm, height 5 cm

2 base 16 mm, height 14 mm

5 base 2.4 m, height 1.5 m

3 base 9 m, height 12 m

6 base 50 cm, height 1 m (in  $\text{cm}^2$ )

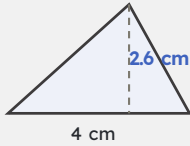
Level 2 · Building up

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

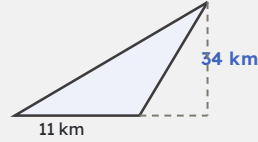
**Example 2** Find the area. The height may be outside.

**a** Find the area.



Your teacher works this out. Copy every step.

**b** Find the area.



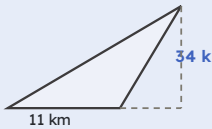
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the area. The height may be outside the triangle.

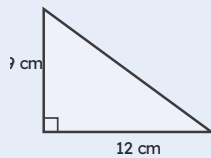
6 QUESTIONS

**REMEMBER** The height is perpendicular (at  $90^\circ$ ) to the base.

1



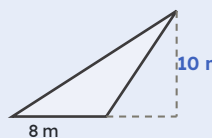
3



5

A triangle has area  $20 \text{ m}^2$  and height 8 m. Find its base.

2



4

A triangle has area  $36 \text{ cm}^2$  and base 9 cm. Find its height.

6

A right-angled triangle has shorter sides 5 cm and 12 cm. Find its area.

**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the height, then explain.

- a** A triangle has area  $36 \text{ cm}^2$  and base  $9 \text{ cm}$ . Find its height.

Your teacher works this out. Copy every step.

- b** Why is the area of a triangle half of a rectangle?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** The height may be outside the triangle.

- 1** A triangular sail has base  $3.2 \text{ m}$  and height  $5 \text{ m}$ . Find its area.

- 3** Find the area of a square of side  $10 \text{ cm}$  with one corner triangle (legs  $4 \text{ cm}$ ) cut off.

- 2** A triangular garden bed has base  $6 \text{ m}$  and height  $4 \text{ m}$ . Mulch covers  $3 \text{ m}^2$  per bag. How many bags?

- 4** Two triangles have the same base  $8 \text{ cm}$ . One has height  $5 \text{ cm}$ , one  $10 \text{ cm}$ . Compare their areas.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Find the area of a kite with diagonals 10 cm and 16 cm.

1. The long diagonal splits it into 2 triangles, base 16, height 5
2.  $2 \times \frac{1}{2} \times 16 \times 5$

Answer: 80 cm<sup>2</sup>

YOU DO

On your own: read the worked example, then try these.

- E1 Find the area of a rhombus with diagonals 6 m and 9 m.

- E2 A triangle has vertices (0, 0), (8, 0) and (3, 6) on a grid. Find its area.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

Level 1

Find the area: base 10 m, height 7 m.

Level 2

Find the area: base 5 cm, height 3.2 cm.

Level 3

A triangle has area 24 cm<sup>2</sup> and base 6 cm. Find its height.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1.  $A = \frac{1}{2} \times \text{base} \times \text{height}$ .
  2. The height is perpendicular (at  $90^\circ$ ) to the base.
  3. The height may be outside the triangle.
  4. Answer in square units.

### Level 1

base 30, height 24

1.  $\frac{1}{2} \times 30 \times 24$

**Answer:**  $360 \text{ mm}^2$

### Level 2

base 4, height 2.6

1.  $\frac{1}{2} \times 4 \times 2.6$

**Answer:**  $5.2 \text{ cm}^2$

### Level 3

Area 36, base 9

1.  $\frac{1}{2} \times 9 \times h = 36$

**Answer:**  $h = 8 \text{ cm}$

### Extension

Kite 10 by 16

1.  $2 \times \frac{1}{2} \times 16 \times 5$

**Answer:**  $80 \text{ cm}^2$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
**8.07**

# Area of a parallelogram

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the area of a parallelogram

## Success criteria

I can:

- ☐ identify the base and perpendicular height
- ☐ use  $A = b \times h$
- ☐ not use the slanted side as the height

**Key terms** Write the definition as your teacher explains it.

**Parallelogram**

**Perpendicular height**

Notes

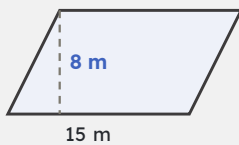
## Level 1 · Getting started

Easy · 18 questions

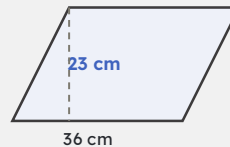
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the area.  $A = bh$ .

**a** Find the area.



**b** Find the area.



Your teacher works this out. Copy every step.

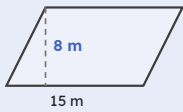
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Find the area.

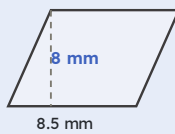
6 QUESTIONS

**REMEMBER**  $A = \text{base} \times \text{height}$ .

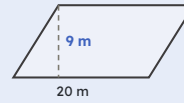
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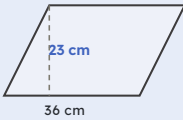

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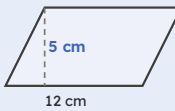

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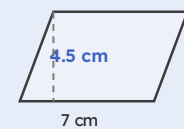

2




4




6




**YOU DO** On your own: **Easy** Set A2. Quick drill: find the area of the parallelogram.

6 QUESTIONS

1  $b = 9 \text{ cm}, h = 4 \text{ cm}$

2  $b = 15 \text{ m}, h = 8 \text{ m}$

3  $b = 6.5 \text{ mm}, h = 2 \text{ mm}$

4  $b = 12 \text{ cm}, h = 12 \text{ cm}$

5  $b = 30 \text{ m}, h = 1.5 \text{ m}$

6  $b = 7 \text{ km}, h = 11 \text{ km}$

**YOU DO** On your own: **Easy** Set B. Find the area of each parallelogram.

6 QUESTIONS

1 base 10 cm, height 6 cm

4 base 8 km, height 3.5 km

2 base 4.5 m, height 2 m

5 base 1.2 m, height 80 cm ( $\text{m}^2$ )

3 base 25 mm, height 12 mm

6 base 30 cm, height 15 cm

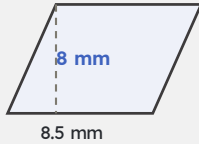
**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Find the area, then explain.

**a** Find the area.



Your teacher works this out. Copy every step.

**b** Why is the area of a parallelogram the same as a rectangle with the same base and height?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the missing value.

6 QUESTIONS

**REMEMBER** Height is perpendicular to the base.

**1** Area  $72 \text{ cm}^2$ , base  $12 \text{ cm}$ . Find the height.

**3** Area  $3.6 \text{ m}^2$ , base  $1.8 \text{ m}$ . Find the height.

**5** Base  $14 \text{ mm}$ , slant side  $9 \text{ mm}$ , height  $7 \text{ mm}$ . Find the area.

**2** Area  $45 \text{ m}^2$ , height  $5 \text{ m}$ . Find the base.

**4** Sides  $10 \text{ cm}$  and  $6 \text{ cm}$ , height  $5 \text{ cm}$  on the  $10 \text{ cm}$  side. Find the area.

**6** A parallelogram and a triangle have base  $8 \text{ cm}$  and height  $6 \text{ cm}$ . Compare their areas.

**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the height, then the area.

- a** A parallelogram has area  $72 \text{ cm}^2$  and base 12 cm. Find its height.

Your teacher works this out. Copy every step.

- b** A parallelogram has sides 10 cm and 6 cm, and height 5 cm on the 10 cm side. Find its area.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Do not use the slanted side.

- 1** A parallelogram-shaped garden has base 12 m and height 7.5 m. Turf costs \$9.80 per  $\text{m}^2$ . Find the cost.

- 3** A rhombus has side 10 cm and height 8 cm. Find its area.

- 2** Draw two different parallelograms with area  $24 \text{ cm}^2$ .

- 4** The same parallelogram has base 12 cm, height 6 cm, and another side 8 cm. Find the height on the 8 cm side.

**Extension · Stretch yourself**

Finished Set D? Stretch yourself here.

**Worked example**

Find the area of a trapezium with parallel sides 8 cm and 14 cm and height 5 cm.

1. Two copies make a parallelogram with base  $8 + 14 = 22$
2. Half of  $22 \times 5$

**Answer:**  $55 \text{ cm}^2$

**YOU DO**

On your own: read the worked example, then try these.

- E1** Find the area of a trapezium with parallel sides 6 m and 10 m and height 4 m.

- E2** A trapezium has area  $60 \text{ cm}^2$ , height 6 cm and one parallel side 8 cm. Find the other.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the area: base 9 cm, height 4 cm.

**Level 2**

Find the area: base 2.5 m, height 1.6 m.

**Level 3**

A parallelogram has area  $50 \text{ m}^2$  and height 4 m. Find its base.

## Lesson summary

Missed the lesson? Start here.

### Remember

1.  $A = \text{base} \times \text{height}$ .
2. Height is perpendicular to the base.
3. Do not use the slanted side.
4. Cut the triangle off one end and move it: a rectangle.

#### Level 1

base 15, height 8

1.  $15 \times 8$

**Answer:**  $120 \text{ m}^2$

#### Level 2

base 8.5, height 8

1.  $8.5 \times 8$

**Answer:**  $68 \text{ mm}^2$

#### Level 3

Area 72, base 12

1.  $72 \div 12$

**Answer:** 6 cm

#### Extension

Trapezium 8, 14, h 5

1.  $\frac{1}{2} \times 22 \times 5$

**Answer:**  $55 \text{ cm}^2$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
**8.08**

# Area of composite shapes

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the area of composite and shaded shapes

## Success criteria

I can:

- ☐ split a shape into rectangles and triangles, then add
- ☐ find a shaded area by subtracting
- ☐ solve practical problems such as painting a wall

**Key terms** Write the definition as your teacher explains it.

**Composite shape**

**Shaded area**

Notes

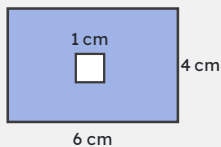
## Level 1 · Getting started

Easy · 10 questions

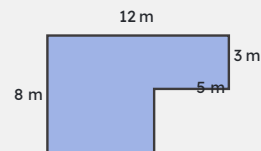
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the area. Add or subtract rectangles.

**a** Find the shaded area.



**b** Find the area.



Your teacher works this out. Copy every step.

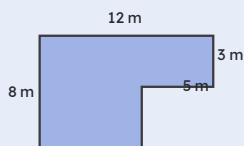
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Find the area by splitting into rectangles.

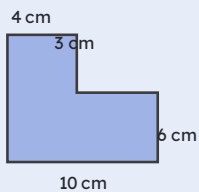
3 QUESTIONS

**REMEMBER** Split the shape into rectangles and triangles.

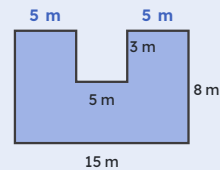
1



2



3



**YOU DO** On your own: **Easy** Set A2. Quick drill: find the area.

4 QUESTIONS

1 a  $10 \times 6$  rectangle with a  $2 \times 2$  square hole (m)

2 two rectangles,  $5 \times 3$  and  $4 \times 2$  (cm)

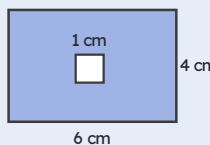
3 a  $8 \times 8$  square minus a  $3 \times 4$  corner (mm)

4 a  $6 \times 4$  rectangle plus a triangle, base 6, height 2 (m)

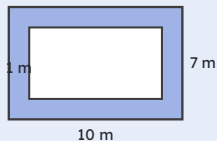
**YOU DO** On your own: **Easy** Set B. Find the shaded area.

3 QUESTIONS

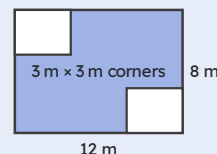
1



2



3



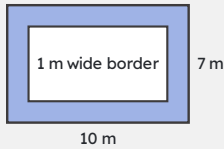
Level 2 · Building up

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

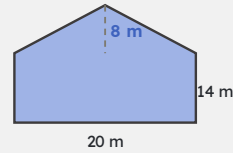
**Example 2** Find the area.

**a** Find the shaded area.



Your teacher works this out. Copy every step.

**b** Find the area.



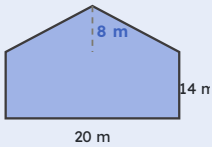
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the area.

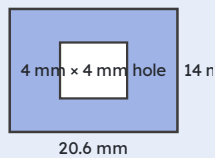
6 QUESTIONS

**REMEMBER** Find missing lengths first.

1



3



5

A 10 m by 6 m rectangle with a triangle of base 10 m and height 4 m on top.

2



4

A rectangle 12 cm by 8 cm with a 4 cm by 3 cm rectangle cut from one corner.

6

A square of side 8 cm with a square of side 3 cm cut from the centre.

**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Solve the problem.

- a** A wall 6.6 m by 3 m has a 2 m by 1 m window and a 0.9 m by 2 m door. One litre of paint covers  $3 \text{ m}^2$ . How many whole litres are needed?

Your teacher works this out. Copy every step.

- b** Find the area of a square of side 5 mm with a triangle of base 5 mm and height 5 mm on top.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Add the parts, or subtract holes (shaded area).

- 1** A wall 6.6 m by 3 m has a 2 m by 1 m window and a 0.9 m by 2 m door. One litre of paint covers  $3 \text{ m}^2$ . How many whole litres are needed?

- 3** Find the area of an L-shaped room: 6 m by 4 m with a 2 m by 1.5 m corner cut out.

- 2** A 50 cm by 35 cm poster has a 40 cm by 28 cm photo on it. Find the area of the poster not covered.

- 4** Carpet costs \$38 per  $\text{m}^2$ . Find the cost of carpeting the room in Question 3.

**Extension · Stretch yourself**

Finished Set D? Stretch yourself here.

**Worked example**

A path 1 m wide runs along two sides of a 12 m by 8 m lawn (an L-shape). Find the area of the path.

1. Outer rectangle  $13 \times 9 = 117 \text{ m}^2$
2. Lawn  $12 \times 8 = 96 \text{ m}^2$
3.  $117 - 96$

**Answer:**  $21 \text{ m}^2$

**YOU DO**

On your own: read the worked example, then try these.

- E1** A 1.5 m wide path goes around all four sides of a 10 m by 6 m pool. Find its area.

- E2** Find the area of a cross made of five 3 cm squares.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the area of a 10 cm × 5 cm rectangle with a 2 cm × 2 cm square hole.

**Level 2**

Find the area of an L-shape made from a 6 m × 3 m and a 2 m × 2 m rectangle.

**Level 3**

A wall is 4 m × 2.5 m with a 1 m × 1 m window. Paint covers 3 m<sup>2</sup> per litre. How many whole litres?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Split the shape into rectangles and triangles.
  2. Find missing lengths first.
  3. Add the parts, or subtract holes (shaded area).
  4. Check units and give square units.

### Level 1

$6 \times 4$  rectangle with  $1 \times 1$  hole

1.  $24 - 1$

**Answer:**  $23 \text{ cm}^2$

### Level 2

L-shape  $12 \times 8$  minus  $5 \times 5$

1.  $96 - 25$

**Answer:**  $71 \text{ m}^2$

### Level 3

Border 1 m wide on  $10 \times 7$

1.  $70 - 8 \times 5$

**Answer:**  $30 \text{ m}^2$

### Extension

Paint the wall

1.  $19.8 - 2 - 1.8 = 16 \text{ m}^2$

2.  $16 \div 3 = 5.3$

**Answer:** 6 L

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
**8.09**

# Prisms, cross-sections and views

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- identify prisms and their cross-sections
- draw the front, side and top views of a solid

## Success criteria

I can:

- ☐ know a prism has the same cross-section all the way through
- ☐ draw a cross-section parallel to the ends
- ☐ sketch front, side and top views

**Key terms** Write the definition as your teacher explains it.

**Prism**

**Cross-section**

**View**

Notes

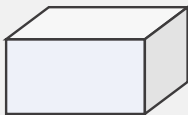
## Level 1 · Getting started

Easy · 15 questions

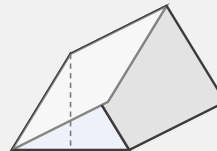
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Is it a prism? Draw its cross-section.

**a** Is it a prism? Draw its cross-section.



**b** Is it a prism? Draw its cross-section.



Your teacher works this out. Copy every step.

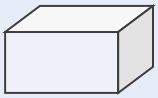
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Is each solid a prism? If so, draw its cross-section.

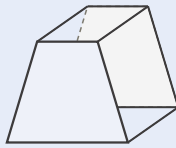
6 QUESTIONS

**REMEMBER** A **PRISM** has two identical, parallel ends joined by rectangles.

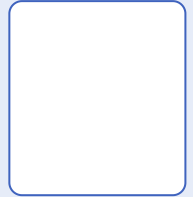
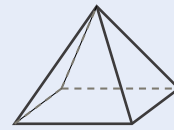
1



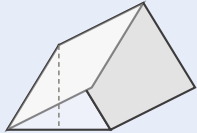
3



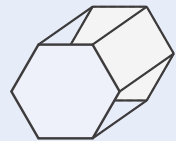
5 a square pyramid



2



4



6 a cylinder



**YOU DO** On your own: **Easy** Set A2. Quick drill: name the shape of the cross-section.

6 QUESTIONS

1 cube

2 triangular prism

3 hexagonal prism

4 rectangular prism

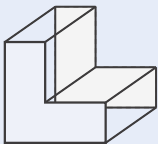
5 cylinder

6 pentagonal prism

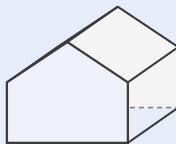
**YOU DO** On your own: **Easy** Set B. Draw the cross-section of each prism.

3 QUESTIONS

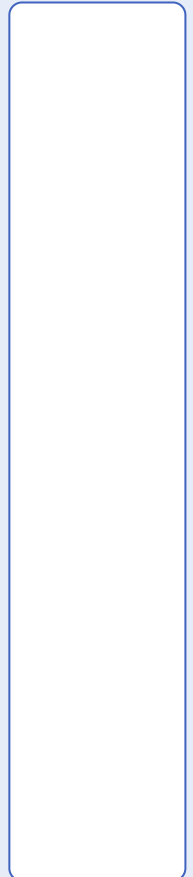
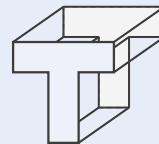
1



2



3



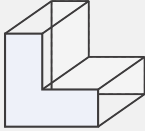
**Level 2 · Building up**

Medium · 2 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Draw the cross-section, then explain.

**a** Draw the cross-section.



Your teacher works this out. Copy every step.

**b** Is a cone a prism? Explain.

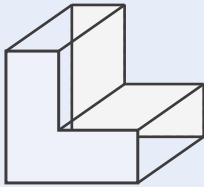
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Sketch the front (F), side and top views.

2 QUESTIONS

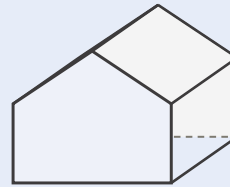
**REMEMBER** The **CROSS-SECTION** is the same shape all the way through.

**1** front, left and top views



F

**2** right, front and top views



F

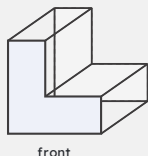
Level 3 · Challenge

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

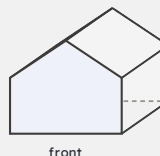
**Example 3** Sketch the three views.

**a** Sketch the front, left and top views.



Your teacher works this out. Copy every step.

**b** Sketch the right, front and top views.



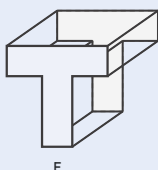
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Views and solids.

4 QUESTIONS

**REMEMBER** Views: look from the front, the side and the top.

**1** Sketch the front, side and top views.



**3** Name a solid whose front, side and top views are all the same.

**2** A solid has a square front view, a square side view and a circle top view. What is it?

**4** Draw the top view of a cylinder lying on its side.



## Extension · Stretch yourself

Finished Set D? Stretch yourself here.



## Worked example

A solid has front view a rectangle, side view a triangle and top view a rectangle. What is it?

1. A triangle from the side means triangular ends
2. Rectangles from the front and top: it is lying on a rectangular face

**Answer:** a triangular prism

## YOU DO

On your own: read the worked example, then try these.

- E1 A solid has front and side views both triangles and top view a square. What is it?

- E2 Draw the front view of a stack of cubes: 3 in the back row, 2 in the middle, 1 at the front, all in a line.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

## YOU DO

## Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

## Level 1

Is a cylinder a prism? Explain.

## Level 2

Name the cross-section of a hexagonal prism.

## Level 3

Sketch the top view of a triangular prism lying on its rectangular face.

## Lesson summary

Missed the lesson? Start here.

### Remember

1. A **prism** has two identical, parallel ends joined by rectangles.
2. The **cross-section** is the same shape all the way through.
3. Views: look from the front, the side and the top.
4. Pyramids, cones and spheres are not prisms.

#### Level 1

Rectangular box

1. Same rectangle all through

**Answer:** prism, rectangle cross-section

#### Level 2

Cone

1. Cross-section shrinks

**Answer:** not a prism

#### Level 3

L-shaped prism

1. Front: an L

**Answer:** L-shape, rectangle, rectangle

#### Extension

Rectangle, triangle, rectangle

1. Triangular ends

**Answer:** triangular prism

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
8.10

# Volume units

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- convert between units of volume

## Success criteria

I can:

- ☐ know  $1 \text{ cm}^3 = 1000 \text{ mm}^3$  and  $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$
- ☐ multiply to a smaller unit, divide to a larger unit
- ☐ count cubes to find volume

**Key terms** Write the definition as your teacher explains it.

**Volume**

**Cubic unit**

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Explain, then convert.

**a** Why is  $1 \text{ cm}^3 = 1000 \text{ mm}^3$ ?

Your teacher works this out. Copy every step.

**b**  $20 \text{ cm}^3 = \_\_\_ \text{ mm}^3$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Convert.

12 QUESTIONS

**REMEMBER**  $1 \text{ cm}^3 = 10 \times 10 \times 10 = 1000 \text{ mm}^3$ .

**1**  $20 \text{ cm}^3$  to  $\text{mm}^3$

**2**  $3 \text{ cm}^3$  to  $\text{mm}^3$

**3**  $2 \text{ m}^3$  to  $\text{cm}^3$

**4**  $0.5 \text{ m}^3$  to  $\text{cm}^3$

**5**  $4.5 \text{ cm}^3$  to  $\text{mm}^3$

**6**  $1.2 \text{ m}^3$  to  $\text{cm}^3$

**Questions 7–12: convert.**

**7**  $4 \text{ cm}^3$  to  $\text{mm}^3$

**8**  $2000 \text{ mm}^3$  to  $\text{cm}^3$

**9**  $1 \text{ m}^3$  to  $\text{cm}^3$

**10**  $500\,000 \text{ cm}^3$  to  $\text{m}^3$

**11**  $0.3 \text{ cm}^3$  to  $\text{mm}^3$

**12**  $2.5 \text{ m}^3$  to  $\text{cm}^3$

**YOU DO** On your own: **Easy** Set B. Convert.

6 QUESTIONS

1  $7500 \text{ mm}^3$  to  $\text{cm}^3$

4  $250\,000 \text{ cm}^3$  to  $\text{m}^3$

2  $4000 \text{ mm}^3$  to  $\text{cm}^3$

5  $600 \text{ mm}^3$  to  $\text{cm}^3$

3  $3\,000\,000 \text{ cm}^3$  to  $\text{m}^3$

6  $80\,000 \text{ cm}^3$  to  $\text{m}^3$

**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Convert.

a  $0.5 \text{ m}^3 = \_\_\_\_ \text{ cm}^3$

Your teacher works this out. Copy every step.

b  $7500 \text{ mm}^3 = \_\_\_\_ \text{ cm}^3$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Convert. Show your working.

6 QUESTIONS

**REMEMBER**  $1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$ .

1  $230\,000 \text{ mm}^3$  to  $\text{m}^3$

3  $5 \text{ km}^3$  to  $\text{m}^3$

5  $12\,500 \text{ cm}^3$  to  $\text{m}^3$

2  $0.004 \text{ m}^3$  to  $\text{mm}^3$

4  $0.75 \text{ m}^3$  to  $\text{cm}^3$

6  $9 \text{ cm}^3$  to  $\text{m}^3$

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Convert, then count cubes.

**a**  $230\,000\text{ mm}^3 = \_\_\_\text{ m}^3$

Your teacher works this out. Copy every step.

**b** How many 1 cm cubes fit in a 5 cm by 4 cm by 3 cm box?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Cubes and volume.

4 QUESTIONS

**REMEMBER** To a smaller unit:  $\times$ . To a larger unit:  $\div$ .

**1** How many 1 cm cubes fit in a box 5 cm by 4 cm by 3 cm?

**3** Order:  $0.02\text{ m}^3$ ,  $15\,000\text{ cm}^3$ ,  $25\,000\,000\text{ mm}^3$ .

**2** How many 2 cm cubes fit in a box 10 cm by 6 cm by 4 cm?

**4** Which unit for: a sugar cube, a swimming pool, a lunch box?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Convert  $6\text{ cm}^3$  to  $\text{mm}^3$ .

**Level 2**

Convert  $2\,500\,000\text{ cm}^3$  to  $\text{m}^3$ .

**Level 3**

Convert  $1500\text{ mm}^3$  to  $\text{cm}^3$ .

**Extension · Stretch yourself****Worked example**How many  $\text{mm}^3$  in  $1 \text{ m}^3$ ?

1.  $1 \text{ m} = 1000 \text{ mm}$
2.  $1000 \times 1000 \times 1000$

**Answer:**  $1\,000\,000\,000 \text{ mm}^3$ **YOU DO**

On your own: read the worked example, then try these.

- E1** Convert  $0.000\,35 \text{ m}^3$  to  $\text{mm}^3$ .

- E2** A cube has volume  $1 \text{ m}^3$ . How many  $10 \text{ cm}$  cubes fit inside it?

**Lesson summary**

Missed the lesson? Start here.

- Volume conversions**
1.  $1 \text{ cm}^3 = 10 \times 10 \times 10 = 1000 \text{ mm}^3$ .
  2.  $1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$ .
  3. To a smaller unit:  $\times$ . To a larger unit:  $\div$ .
  4. Volume is measured in cubic units.

**Level 1** $20 \text{ cm}^3$  to  $\text{mm}^3$ 

$$1. \times 1000$$

**Answer:**  $20\,000 \text{ mm}^3$ **Level 2** $0.5 \text{ m}^3$  to  $\text{cm}^3$ 

$$1. \times 1\,000\,000$$

**Answer:**  $500\,000 \text{ cm}^3$ **Level 3** $230\,000 \text{ mm}^3$  to  $\text{m}^3$ 

$$1. \div 1\,000\,000\,000$$

**Answer:**  $0.00023 \text{ m}^3$ **Extension** $1 \text{ m}^3$  in  $\text{mm}^3$ 

$$1. 1000^3$$

**Answer:**  $1\,000\,000\,000$ 

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LESSON  
**8.11**

# Volume of a rectangular prism

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the volume of rectangular prisms

## Success criteria

I can:

- ☐ use  $V = l \times w \times h$
- ☐ use  $V = \text{area of base} \times \text{height}$
- ☐ find a missing dimension

**Key terms** Write the definition as your teacher explains it.

**Volume**

**Base area**

Notes

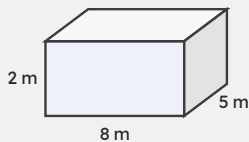
## Level 1 · Getting started

Easy · 18 questions

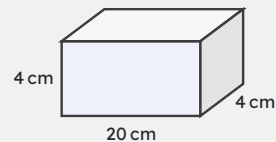
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the volume.  $V = lwh$ .

**a** Find the volume.



**b** Find the volume.



Your teacher works this out. Copy every step.

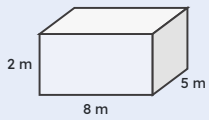
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Find the volume.

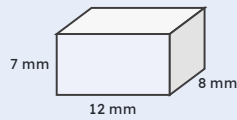
6 QUESTIONS

**REMEMBER**  $V = \text{length} \times \text{width} \times \text{height}$ .

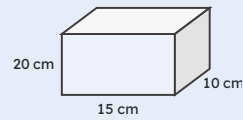
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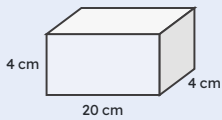

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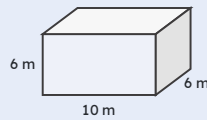

5



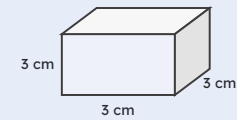

2




4




6




**YOU DO** On your own: **Easy** Set A2. Quick drill: find the volume.

6 QUESTIONS

1  $5 \text{ cm} \times 4 \text{ cm} \times 2 \text{ cm}$

2 cube, edge 3 m

3  $10 \text{ mm} \times 6 \text{ mm} \times 5 \text{ mm}$

4  $8 \text{ m} \times 2 \text{ m} \times 1.5 \text{ m}$

5 cube, edge 10 cm

6  $12 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$

**YOU DO** On your own: **Easy** Set B. Find the volume.

6 QUESTIONS

1  $6 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$

4  $10 \text{ mm} \times 10 \text{ mm} \times 3 \text{ mm}$

2 cube with edge 5 m

5 base area  $24 \text{ cm}^2$ , height 5 cm

3  $2.5 \text{ m} \times 2 \text{ m} \times 1 \text{ m}$

6  $1.2 \text{ m} \times 0.5 \text{ m} \times 0.5 \text{ m}$

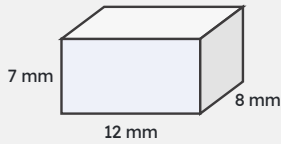
**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

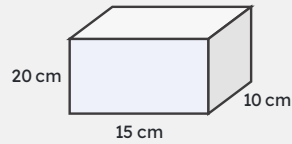
**Example 2** Find the volume.

**a** Find the volume.



Your teacher works this out. Copy every step.

**b** Find the volume.



Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the missing value.

6 QUESTIONS

**REMEMBER** Or  $V = \text{area of base} \times \text{height}$ .

**1** A cube has volume  $64 \text{ cm}^3$ . Find its edge.

**3** Volume  $90 \text{ m}^3$ , base area  $15 \text{ m}^2$ . Find the height.

**5** A box is 40 cm by 30 cm by 25 cm. Find its volume in  $\text{m}^3$ .

**2** Volume  $120 \text{ cm}^3$ , length 6 cm, width 4 cm. Find the height.

**4** A cube has volume  $1000 \text{ mm}^3$ . Find its edge.

**6** Double every edge of a 2 cm cube. How many times bigger is the volume?

**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the edge or the height.

**a** A cube has volume  $64 \text{ cm}^3$ . Find its edge length.

Your teacher works this out. Copy every step.

**b** A box has volume  $120 \text{ cm}^3$ , length 6 cm and width 4 cm. Find its height.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Cube:  $V = s^3$ .

**1** A shipping container is 12 m by 2.4 m by 2.6 m. Find its volume.

**3** Sand is piled into a trailer 2 m by 1.2 m by 0.4 m. Find the volume of sand.

**2** How many 5 cm cubes fit in a box 30 cm by 20 cm by 15 cm?

**4** A rectangular prism has volume  $48 \text{ cm}^3$ . Find 3 possible sets of whole-number dimensions.

**Extension · Stretch yourself**

Finished Set D? Stretch yourself here.

**Worked example**

An L-shaped prism has a cross-section of area  $90 \text{ m}^2$  and length  $7 \text{ m}$ . Find its volume.

1.  $V = \text{area of cross-section} \times \text{length}$
2.  $90 \times 7$

**Answer:**  $630 \text{ m}^3$

**YOU DO**

On your own: read the worked example, then try these.

- E1** A prism has an L-shaped cross-section of area  $45 \text{ cm}^2$  and length  $12 \text{ cm}$ . Find its volume.

- E2** A room is  $5 \text{ m}$  by  $4 \text{ m}$  by  $2.7 \text{ m}$ . An air conditioner cycles  $1 \text{ m}^3$  of air every  $2$  seconds. How long to cycle all the air?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO**

**Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the volume:  $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$ .

**Level 2**

Find the volume of a cube with edge  $6 \text{ m}$ .

**Level 3**

A box has volume  $72 \text{ cm}^3$  and base area  $12 \text{ cm}^2$ . Find its height.

## Lesson summary

Missed the lesson? Start here.

**Remember** 1.  $V = \text{length} \times \text{width} \times \text{height}$ .  
3. Cube:  $V = s^3$ .

2. Or  $V = \text{area of base} \times \text{height}$ .  
4. Answer in cubic units.

### Level 1

$$8 \times 5 \times 2$$

1.  $40 \times 2$

Answer:  $80 \text{ m}^3$

### Level 2

$$15 \times 10 \times 20$$

1.  $150 \times 20$

Answer:  $3000 \text{ cm}^3$

### Level 3

$$\text{Cube } 64 \text{ cm}^3$$

1.  $4 \times 4 \times 4 = 64$

Answer: 4 cm

### Extension

L-prism: area 90, length 7

1.  $90 \times 7$

Answer:  $630 \text{ m}^3$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
**8.12**

# Volume of a prism

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the volume of prisms with triangular and other cross-sections

## Success criteria

I can:

- ☐ find the area of the cross-section first
- ☐ use  $V = A \times h$  ( $h$  is the length of the prism)
- ☐ use this for triangular, trapezium and composite prisms

**Key terms** Write the definition as your teacher explains it.

**Cross-section**

**Prism**

Notes

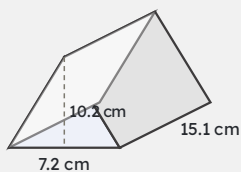
## Level 1 · Getting started

Easy · 13 questions

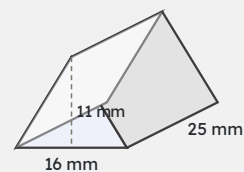
**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the volume.  $V = Ah$ .

**a** Find the volume.



**b** Find the volume.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Easy** Set A. Questions 1–6: Find the volume. ( $A$  = cross-section area,  $h$  = length)

10 QUESTIONS

**REMEMBER** Identify the cross-section (the end face).

1  $A = 20 \text{ cm}^2$ ,  $h = 5 \text{ cm}$

2  $A = 35 \text{ cm}^2$ ,  $h = 8 \text{ cm}$

3  $A = 12 \text{ m}^2$ ,  $h = 3 \text{ m}$

4  $A = 7.5 \text{ mm}^2$ ,  $h = 4 \text{ mm}$

5  $A = 100 \text{ cm}^2$ ,  $h = 0.5 \text{ cm}$

6  $A = 48 \text{ m}^2$ ,  $h = 10 \text{ m}$

Questions 7–10: find the volume ( $A$  = area of cross-section,  $h$  = length).

7  $A = 15 \text{ cm}^2$ ,  $h = 4 \text{ cm}$

8  $A = 9 \text{ m}^2$ ,  $h = 7 \text{ m}$

9  $A = 2.5 \text{ cm}^2$ ,  $h = 10 \text{ cm}$

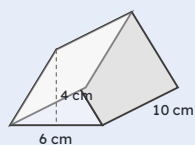
10  $A = 40 \text{ mm}^2$ ,  $h = 3 \text{ mm}$

**YOU DO**

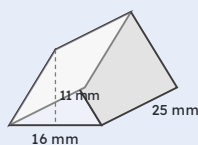
On your own: **Easy** Set B. Find the area of the triangular cross-section, then the volume.

3 QUESTIONS

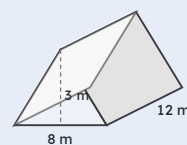
1



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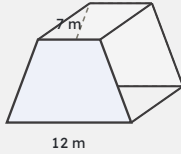
**Level 2 · Building up**

Medium · 3 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Find the volume. Find the area of the cross-section first.

- a** Find the volume (trapezium cross-section: parallel sides 7 m and 12 m, height 6 m, length 10 m).



Your teacher works this out. Copy every step.

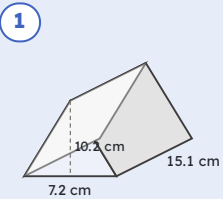
- b** A prism has cross-section area  $35 \text{ cm}^2$  and length 8 cm. Find its volume.

Your teacher works this out. Copy every step.

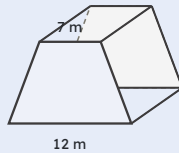
**YOU DO** On your own: **Medium** Set C. Find the volume.

3 QUESTIONS

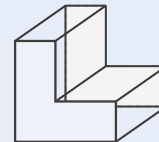
**REMEMBER** Find its area A.



- 2** trapezium: 7 m and 12 m, height 6 m; length 10 m



- 3** L-shape area  $45 \text{ cm}^2$ , length 12 cm



**Level 3 · Challenge**

Challenging · 4 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the volume or the cross-section.

- a** Find the volume of a triangular prism: triangle base 6 cm, height 4 cm, prism length 10 cm.

Your teacher works this out. Copy every step.

- b** A prism has volume  $360 \text{ m}^3$  and length 12 m. Find the area of its cross-section.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER**  $V = A \times \text{length of the prism.}$

- 1** A tent is a triangular prism: triangle base 2.4 m, height 1.5 m, length 3 m. Find its volume.

- 3** A prism has volume  $360 \text{ m}^3$  and length 12 m. Find the cross-section area.

- 2** A ramp has a right-triangle cross-section with legs 1.2 m and 0.5 m and width 2 m. Find its volume.

- 4** A water trough has a trapezium cross-section (parallel sides 60 cm and 40 cm, depth 30 cm) and length 2 m. Find its volume in  $\text{cm}^3$ .

## Extension · Stretch yourself

Finished Set D? Stretch yourself here.

## Worked example

A swimming pool is 25 m long and 10 m wide. It is 1 m deep at one end and 2 m deep at the other. Find its volume.

1. Cross-section (side view) is a trapezium:  $\frac{1}{2} \times (1 + 2) \times 25 = 37.5 \text{ m}^2$
2. Width 10 m:  $37.5 \times 10$

Answer:  $375 \text{ m}^3$ 

## YOU DO

On your own: read the worked example, then try these.

- E1 A pool 50 m by 20 m is 1.2 m deep at one end and 2 m at the other. Find its volume.

- E2 How many litres is the pool in the worked example? ( $1 \text{ m}^3 = 1000 \text{ L}$ )

✕ Show off your skill: tear along this line and hand it to your teacher ✕

## YOU DO

## Show off your skill

Choose ONE question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

## Level 1

A prism has cross-section  $15 \text{ cm}^2$  and length 6 cm. Find its volume.

## Level 2

Triangle base 5 m, height 4 m, prism length 9 m. Find the volume.

## Level 3

A prism has volume  $200 \text{ cm}^3$  and cross-section  $25 \text{ cm}^2$ . Find its length.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Identify the cross-section (the end face).
  2. Find its area  $A$ .
  3.  $V = A \times \text{length of the prism}$ .
  4. Triangle:  $A = \frac{1}{2}bh$ . Trapezium:  $A = \frac{1}{2}(a + b)h$ .

### Level 1

Triangle  $7.2 \times 10.2$ ,  
length 15.1

1.  $A = \frac{1}{2} \times 7.2 \times 10.2 = 36.72$
2.  $\times 15.1$

**Answer:**  $554.472 \text{ cm}^3$

### Level 2

$A = 35$ ,  $h = 8$

1.  $35 \times 8$

**Answer:**  $280 \text{ cm}^3$

### Level 3

Trapezium 7, 12, height  
6, length 10

1.  $\frac{1}{2} \times 19 \times 6 = 57$
2.  $\times 10$

**Answer:**  $570 \text{ m}^3$

### Extension

Sloping pool

1.  $37.5 \times 10$

**Answer:**  $375 \text{ m}^3$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
8.13

## Capacity

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the capacity of a container in litres

## Success criteria

I can:

- ☐ know  $1 \text{ cm}^3 = 1 \text{ mL}$  and  $1 \text{ m}^3 = 1000 \text{ L}$
- ☐ find volume, then convert to litres
- ☐ solve problems about water levels and large volumes

**Key terms** Write the definition as your teacher explains it.

Capacity

Megalitre

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Find the capacity, then the water height.

- a** A box is 30 cm by 20 cm by 10 cm. Find its capacity in litres.

Your teacher works this out. Copy every step.

- b** The water is poured into a prism with base 20 cm by 20 cm. How high does it reach?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Convert.

12 QUESTIONS

**REMEMBER**  $1 \text{ cm}^3 = 1 \text{ mL}$ ;  $1000 \text{ cm}^3 = 1 \text{ L}$ .

1  $500 \text{ cm}^3$  to mL

2  $2000 \text{ cm}^3$  to L

3  $3.5 \text{ m}^3$  to L

4  $0.2 \text{ m}^3$  to L

5 4500 L to kL

6 2 ML to L

Questions 7–12: convert.

7  $1000 \text{ cm}^3$  to L

8  $250 \text{ cm}^3$  to mL

9  $2 \text{ m}^3$  to L

10 3500 mL to L

11 6 kL to L

12  $0.5 \text{ m}^3$  to L

**YOU DO** On your own: **Easy** Set B. Find the capacity in litres.

6 QUESTIONS

1 30 cm × 20 cm × 10 cm

4 50 cm × 40 cm × 30 cm

2 8 m × 5 m × 2 m

5 1 m × 1 m × 1 m

3 20 cm × 4 cm × 4 cm

6 10 cm × 10 cm × 10 cm

**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Convert, then find the capacity.

a Convert 3.5 m<sup>3</sup> to litres.

Your teacher works this out. Copy every step.

b Find the capacity in litres of a tank 8 m by 5 m by 2 m.

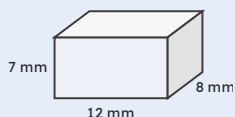
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the capacity in litres of each prism.

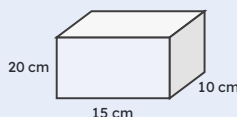
6 QUESTIONS

**REMEMBER** 1 m<sup>3</sup> = 1000 L = 1 kL.

1



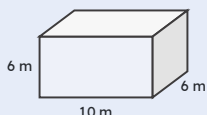
3



5

trapezium prism: 570 m<sup>3</sup>

2



4

triangular prism:  $\frac{1}{2} \times 16 \times 11$  mm, length 25 mm

6

cube, edge 2.5 m

Level 3 · Challenge

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Solve.

- a** An iceberg is like a prism 160 km by 50 km by 250 m. How many megalitres of water?

Your teacher works this out. Copy every step.

- b** A 2 L jug fills cups of 250 mL. How many cups?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** 1 ML (megalitre) = 1 000 000 L.

- 1** A 30 cm × 20 cm × 10 cm box of water is poured into a prism with base 20 cm × 20 cm. How high does the water reach?

- 3** A fish tank 60 cm × 30 cm × 40 cm is filled to 5 cm from the top. How many litres of water?

- 2** Iceberg B9 was like a prism 160 km by 50 km by 250 m. How many megalitres of water did it hold?

- 4** A 1.5 m by 0.8 m bath fills at 12 L a minute. How deep is the water after 10 minutes (cm)?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Convert 2500 cm<sup>3</sup> to litres.

**Level 2**

Find the capacity in litres of a box 25 cm × 20 cm × 10 cm.

**Level 3**

Convert 0.6 m<sup>3</sup> to litres.

**Extension · Stretch yourself****Worked example**

Rain of 10 mm falls on a 12 m by 8 m roof. How many litres run into the tank?

1.  $10 \text{ mm} = 0.01 \text{ m}$
2.  $V = 12 \times 8 \times 0.01 = 0.96 \text{ m}^3$
3.  $\times 1000$

Answer: 960 L

**YOU DO**

On your own: read the worked example, then try these.

- E1** How much rain (mm) must fall on the roof to fill a 4800 L tank?

- E2** A house uses 180 L per person per day. How many days does 960 L last a family of 4?

**Lesson summary**

Missed the lesson? Start here.

**Capacity**

1.  $1 \text{ cm}^3 = 1 \text{ mL}$ ;  $1000 \text{ cm}^3 = 1 \text{ L}$ .

3.  $1 \text{ ML (megalitre)} = 1\,000\,000 \text{ L}$ .

2.  $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$ .

4. Find the volume first, then convert.

**Level 1**

$30 \times 20 \times 10 \text{ cm}$

1.  $6000 \text{ cm}^3$

Answer: 6 L

**Level 2**

Pour into  $20 \times 20$  base

1.  $6000 \div 400$

Answer: 15 cm (D)

**Level 3**

$8 \times 5 \times 2 \text{ m}$

1.  $80 \text{ m}^3 \times 1000$

Answer: 80 000 L

**Extension**

Rain on a roof

1.  $0.96 \text{ m}^3$

Answer: 960 L

This space is left blank on purpose. It is the back of your exit ticket.